

OP Sales Analytics — Individual Project

Course: Data Analytics in Sales Management · 2025/2026

0) Objective

Turn OriginalPeople's raw data into a clean analytical model, extract actionable sales insights, and propose concrete next steps (plus a simple forecast). You may use any mix of Power BI, Tableau, Excel, Python, and SQL. You will submit your cleaned dataset(s), analysis (dashboard or notebook), and a concise write-up.

Data sources (CSV and/or DB access): orders (with JSON field basket), app_clicks (website events), dm_clicks (paid media clicks; a separate CSV with costs/ROAS per ad will be provided), and abandonedbasket. Table ages vary—assess each table's time coverage and take that into account in your analysis.

1) Data Normalization & Structuring (MANDATORY)

Deliver a short Data Prep Report (min 2 pages) that explains your steps and choices.

1.1 Nulls & validity checks

- For each table and every field you use (including those extracted from basket), report % NULL/empty and distinct counts.
- Flag outliers and invalid values (e.g., negative amounts, unknown currencies).
- State the rule applied per issue: drop, fix, or keep (with rationale).

1.2 Extract & flatten orders.basket

- Header table/view: order_id, createdate, sessionId, currency, language, ship_country, ship_province, deliveryMethod, order_amount (choose a consistent final amount source such as receiverList[0].amount or the sum of orderItemList).
- Line-items table/view: explode orderItemList → order_id, sku, qty, unit_price, subtotal, description, color. Exclude shipping rows from product analytics.

1.3 Normalize currencies to EUR (MANDATORY)

- Convert all non-EUR orders to EUR and store a final numeric amount_eur.
- Use a documented FX approach (average per currency or historical rates) and cite your source/assumptions.
- Ensure every revenue/AOV/ROAS metric in your analysis uses EUR consistently.

1.4 Keys & joins

- Orders ↔ app_clicks: join on sessionId (exact).
- Orders ↔ dm_clicks: last-touch join on sessionId where click timestamp ≤ order timestamp. If no match, label as “Organic/Direct”.
- If you apply a time window (e.g., last paid click within 7 days), document it clearly.

Deliverables for Section 1: ER diagram or table list of your final model; scripts/notebooks/Power Query steps to reproduce your cleaned tables.

2) KPIs & Analyses (MANDATORY SET)

Build visuals/tables for the following. Each chart must include a one-line takeaway (‘so what?’).

2A) Sales by Area

- Revenue (EUR) and orders by country (and, if available, province/region).
- Include conversion rate (CVR) by country for periods where funnel analysis is applicable.

2B) Sales by SKU / Product Mix

- Top SKUs by units and revenue.
- Attach rate for key add-ons (% of orders that include the add-on).
- Optional: basket composition (avg items/order; add-on uplift to AOV).

2C) Seasonality

- Monthly orders and revenue (EUR) for the maximum period available to you (use a rolling average to smooth).
- Call out peaks (e.g., Q4/Christmas) and low seasons explicitly.

2D) Sales Funnel (clicks → checkout intent → order)

- Join app_clicks, dm_clicks and orders using session ID. Session IDs missing in dm_clicks should be labelled as “organic”.
- Use app_clicks to count sessions (unique sessionid), abandonedbasket (and/or relevant click events) as checkout intent, and orders as purchases.
- Build the funnel as far back in time as your data allows—verify coverage per table and state any gaps.
- Provide counts and step-wise conversion rates. Include a breakdown by country or language.
- Identify exit points using the last event per sessionid in app_clicks (top exit pages/events).

2F) Repeat Buyers

- Define a customer proxy (e.g., anonymized email hash) to distinguish first-time vs repeat orders.
- Report: share of first-time vs repeat orders, and a simple cohort view (first-purchase month vs repeat within 3/6 months).
- If ID quality is limited, state the limitation and proceed with a reasonable proxy.

3) Insights, Strategy & Forecast (MANDATORY)

- Top 5 Insights: each names the segment, observation, and implication.
- Sales Strategy (3–5 actions): target, tactic, expected lift, impact sizing (€), experiment design, and timeline.
- Simple Forecast (3–6 months): seasonal naïve, moving average, exponential smoothing, or a basic time-series model. Include assumptions and a simple uncertainty band or best/base/worst.

4) Deliverables & Format

- Data Prep Report (PDF, 1–2 pages): missingness summary, EUR normalization approach, join rules, and final model diagram.
- Analysis: dashboard (Power BI/Tableau/Excel) or notebook+slides; required visuals with one-line takeaways.
- Strategy Deck (PDF, max 8 slides): executive summary; KPI visuals; 3–5 actions; forecast & assumptions.
- Reproducibility: include SQL/Python/Power Query steps or BI source files, plus a short README on how to run/open.

5) Grading (100 pts)

- Data prep & modeling (25) — clean extractions, EUR normalization, sensible joins, clear documentation.
- KPI accuracy & completeness (35) — all mandatory KPIs, correct definitions, insightful breakdowns.
- Insights & storytelling (20) — clear takeaways; prioritized, non-obvious insights.
- Strategy & forecast (20) — concrete actions with sizing + a reasonable forecast & assumptions.
- Bonus up to +5 for advanced analyses (e.g., product bundling rules, province heatmaps, device effects).

6) Definitions & Hints

- Order Amount (EUR): your consistent final charge field converted to EUR (store as amount_eur).
- AOV: $\text{sum}(\text{amount_eur}) / \text{count}(\text{orders})$.
- CVR (sessions→orders): $\text{orders} / \text{unique sessions}$ (same window).
- Attach Rate (SKU X): $\text{orders containing SKU X} / \text{total orders}$.
- Last-touch attribution: most recent paid click for that session at or before order time; else Organic/Direct.
- Repeat Buyer: same customer proxy ordering on multiple distinct dates.
- Label axes and units clearly; append a one-line takeaway beneath each chart.
- Keep a small 'Assumptions' box (currency method, attribution rule, filters applied).

Note: All tables have varying historical coverage. Assess time coverage for each table and reflect it in your joins, funnel windows, and conclusions.

7) In-class presentation

You will present your sales plan in class as a “sales pitch”. You’ll have a budget of 150k.

The class will be your buyers, in this case your fictional managers, and your goal is to use your data analysis to convince them of your sales plan.